

Jayesh Limaye

More and more people are viewing the PC as a source of entertainment. The television remains the major home entertainer, but for many PC users, nothing beats the convenience of TV right on the PC.

After a hiatus of two long years, we bring to you this, our fresh shootout of TV-Tuners available in India. We take a look at tuners of all shapes and kinds, and will help you decide what suits you best. If you haven't thought about TV on the PC, trust us, once you've tried it, you will not be able to live without it!

There are quite a few players in this category—veterans such as AVerMedia, Compro, PixelView, and Pinnacle, as also new players such as Hauppauge, Leadtek WinFast, and Tech-Com. Of the 25 TV Tuners that arrived at our Test Centre, there were 12 PCI TV-Tuners, nine external USB ones, and four set-top TV-Tuners. We assure you these are the choicest solutions available in India!

INTERNAL TV-TUNERS

The most popular and the oldest kind of TV-Tuners are those that fit inside the PC cabinet—the internal TV-Tuner. These cards fit in a free PCI (or in some cases, PCI-E or AGP) slot of the motherboard. These are the most inexpensive, and are generally richer in features.

Features

The Chipset

The chipset around which a TV-Tuner is built is a basic thing that anyone should note before venturing to purchase it. The chipset determines the absolute highest quality that a TV-Tuner card can deliver. The ageing yet reliable Philips chipset is one of the most popular amongst manufacturers; this is because of its splendid performance thanks to its 9-bit ADC (Analogue to Digital Converter) and its amazingly low price. But there are differences to note here: the Philips SAA713x series can only

deliver mono TV sound, while the rest are capable of stereo. The AVerMedia AVerTV Go 007 FM Plus, Compro VideoMate TV Gold⁺II, and Pinnacle have such chips.

Then there are 10-bit ADC chips from Conexant (the CX2388x series) which also deliver excellent video quality, albeit at a much higher price. Hauppauge, Leadtek Winfast, and PixelView sport such chips, and are all stereo TV sound capable, including the PixelView PlayTV Pro 2, which has the older Conexant Fusion 878A chip.

Hardware MPEG-2 encoding offloads the CPU from the task of MPEG-2 encoding. This is not of much relevance these days, though, because computers have ample horsepower at their disposal. The Leadtek WinFast PVR2000 is the only one that has a Conexant MPEG-2 encoder chip.

FM Radio

All the cards had FM radio, except for the Pinnacle PCTV Stereo and Tech-Com SSD-TV-670. FM Radio is not an expensive implementation, and is a great value-add.

I/O Ports

All the cards have S-Video inputs, while all but the Pinnacle PCTV Pro PCI and PCTV PCI do not have composite inputs. Connect your old camcorder or a VCP to these inputs and you can capture and convert old analogue memories from tapes to digital format on your hard drive—a reason strong enough for some people to buy a TV-Tuner.

The Remote Control

All the TV-Tuners came with remotes. Remote controls come in all shapes and sizes, some as large as police batons (the PixelView PlayTV P3000), while some were card-remotes that will fit in your palm.

Unlike the last time we tested TV-Tuners this time round, the remotes presented better features such as recording and timeshift along with media player controls. Only the Tech-Com SSD-TV-670 has a remote without timeshift.

How We Tested

The test bench comprised an AMD Athlon64 3800+ processor on an MSI RS480M2 motherboard with 1 GB of Corsair DDR RAM running at 400 MHz and a 250 GB 7200 rpm Seagate Barracuda SATA II hard drive. The video card was a GeForce 6600, and the sound card was a Creative SoundBlaster Audigy 2 Platinum. Windows XP SP2 along with the latest versions of all necessary drivers as well as DirectX was installed on the testbed.

We created a System Restore Point, which would serve as a reference point to which we would restore the system after testing each TV-Tuner. This is done in order to avoid possible driver conflict issues: many TV-Tuners that featured the same chip had drivers that consisted of similarly-named files, but which differed in the version number.

We noted the various features of the TV-Tuners such as the chip, presence of a hardware encoder, whether FM radio is supported, and whether the sound was stereo or mono. We noted the number of input/output ports. We also rated the bundled accessories such as a remote control and its functionality, cables, and the software package. Where PVR

software was provided, its functionality—such as whether timeshifting was available and whether MPEG-1/2/4 capture facility was provided—was noted and rated. Windows XP Media Center Edition compatibility and Vista readiness were also noted.

The ease of installation of each device was noted and rated. In the cable TV channel detection test, we deliberately created a cable TV point with weak signal strength. This was achieved by introducing joints in the cable that caused signal strength drops. The TV viewing utility provided with the TV-Tuner was used for determining the number of channels each TV-Tuner detected. (There were a total of 90 cable channels available) We also noted whether channels required fine-tuning.

After this test was done, we rated the visual quality of the detected TV channels on a scale of 10. This was a subjective test, so we looked for artefacts and anomalies such as the feathering effect.

Other issues such as crashing of applications and drivers were also noted. Finally, the features, performance, and price together determined the final scores, and thus, the winners.